

**Machine Learning Based Groundwater Level Predictions as a Proxy for
Compound Flooding Events in Miami, Florida**

Beto Estrada,^a Philippe Tissot,^b Scott A. King,^a Michael C. Sukop,^c Jayantha Obeysekera,^c

^a *Department of Computer Science, Texas A&M University - Corpus Christi, Corpus Christi, TX,
USA*

^b *Conrad Blucher Institute for Surveying and Science, Texas A&M University - Corpus Christi,
Corpus Christi, TX, USA*

^c *Florida International University, 11200 SW 8th Street, OE 148, Miami, FL 33199*

Corresponding author: Beto Estrada, betoejr10@yahoo.com

10 ABSTRACT: Miami, Florida is prone to compound flooding (when multiple flood drivers occur
11 concurrently or within a similar time period) which impacts property and lives. This is in large part
12 due to anthropogenic changes to the region, relative sea level rise, and an increase in precipitation
13 intensity. Better predictive modeling of these events is important to prepare for and mitigate their
14 impact and improve the long-term resiliency of the region and its community. Miami, Florida is
15 home to a karst aquifer that is vulnerable to saltwater intrusion. To combat this, water is pumped
16 into canals throughout the city which helps to recharge the groundwater supply. Water can also be
17 pumped into these canals during extreme precipitation events to decrease the chances or severity of
18 flooding by diverting some of the excess water entering the water conservation areas (WCAs). The
19 important role that groundwater plays in the flooding dynamic of Miami's urban areas suggests
20 that groundwater level may be used as a potential indicator for compound flooding. A 7-year data
21 set (Oct. 2017 - Dec. 2024) including groundwater level, precipitation, ocean water level, and
22 parameters from a nearby spillway serves as the basis for the study. The study confirms that the
23 highest groundwater levels are correlated with flooding events in Miami-Dade County, Florida.
24 Multiple machine learning multilayer perceptron (MLP) models were developed and analyzed that
25 accurately predict the groundwater level at lead times 3, 6, 12, and 24 hours at a target location in
26 Miami-Dade County, Florida. These groundwater level predictions were then used as a proxy for
27 compound flooding events at the target location. The performance of the models were tested using
28 different combinations of inputs at each lead time. A case study was conducted that evaluated
29 the performance of the models during confirmed flooding events. Results show that models that
30 used a combination of past measurements for each feature as well as ocean water level, rainfall,
31 and gate opening perfect prognosis predictions as input performed the best both statistically and
32 during flooding events. The study also quantified the importance of rainfall and its accuracy as
33 a predictor of groundwater level. This study models the different drivers of compound flooding
34 in Miami, Florida, such as groundwater level, surge conditions, and rainfall, using data-driven
35 machine learning methods.

36 SIGNIFICANCE STATEMENT: The purpose of this study is to build toward better predictive
37 modeling of compound flooding events in Miami, Florida, an area that is prone to these events.
38 This is important because compound flooding is a phenomenon where multiple flood drivers occur
39 concurrently or within a similar time period and can cause substantial damage to property and
40 lives. Our results show that using a data-driven method, a machine learning model can be trained
41 to predict groundwater level as a proxy for compound flooding events in Miami, Florida, provided
42 that accurate and reliable predictions for different flood drivers, such as rainfall, are available at
43 the time of inference.

44 1. Introduction

45 Miami, Florida, is prone to compound flooding events (when multiple flood drivers occur
46 concurrently or within a similar time period), which impact property and lives (Wahl et al. 2015).
47 In Miami, rainfall, storm surge, river discharge, and maximum soil moisture can be identified as
48 the main drivers of flooding, with extreme rainfall being the main contributor in the area (Ali et al.
49 2023). Further anthropogenic changes to the region, relative sea level rise, and an increase in
50 precipitation intensity threaten to exacerbate the compound flood risk (Seneviratne et al. 2021).
51 Better predictive modeling of these events is important to prepare for and mitigate their impact
52 and improve the long term resiliency of the region and its community. Miami, Florida sits atop
53 the Biscayne Aquifer which is the main source of freshwater in the area. It is a karst aquifer that
54 is vulnerable to saltwater intrusion. To combat this, water is pumped into canals throughout the
55 city which helps to recharge the groundwater supply. Water can also be pumped into these canals
56 during extreme precipitation events to decrease the chances or severity of flooding by diverting
57 some of the excess water entering the water conservation areas (WCAs) (Czajkowski et al. 2018).
58 This leads to a juggling act between saltwater intrusion management and flood prevention and
59 highlights the role that groundwater level plays in the flooding dynamic of Miami, Florida.

60 a. Objective

61 The main objective of this work was to build toward an operational model that can predict
62 compound flooding events in real-time. The model should be able to provide accurate and timely
63 predictions that alert stakeholders and the general public about potential compound flooding threats

64 for the area. The proposed system follows a data-driven approach which requires fewer inputs and
65 computing resources to make real-time predictions compared to more comprehensive physical
66 models.

67 *b. Approach*

68 The important role that groundwater plays in the flooding dynamic of Miami's urban areas
69 suggests that groundwater may be used as a potential indicator for compound flooding (Sukop et al.
70 2018). This research developed and analyzed machine learning (ML) to predict the groundwater
71 level for a target location in Miami-Dade County, Florida (shown in Fig. 1) using precipitation,
72 ocean water level, parameters from a nearby spillway, and past and current groundwater level
73 measurements as input. The groundwater level predictions were then used as a proxy for compound
74 flooding events at this location.

78 *c. Contributions*

79 The main contributions of this research are:

- 80 1. **Machine learning groundwater level predictions:** Developed and analyzed multilayer per-
81 ceptron (MLP) models that accurately predict groundwater level using multiple inputs, in-
82 cluding precipitation, spillway parameters, ocean water levels, and prior groundwater levels
83 in Miami, Florida.
- 84 2. **Groundwater level as a proxy for compound flooding events:** A case study confirmed that
85 using groundwater level was effective in predicting floods with lead times up to 24 hours for
86 the study area.
- 87 3. **Importance of rainfall predictions as input:** Quantified the importance of precipitation and
88 its accuracy as a predictor of future groundwater level.

89 **2. Related Work**

90 There are multiple studies that focus on flood mitigation and flood warning systems. Physics-
91 based models exist that provide high resolution water level simulations for large watershed grids.
92 There are also ML models that use data-driven methods to predict water levels using less inputs

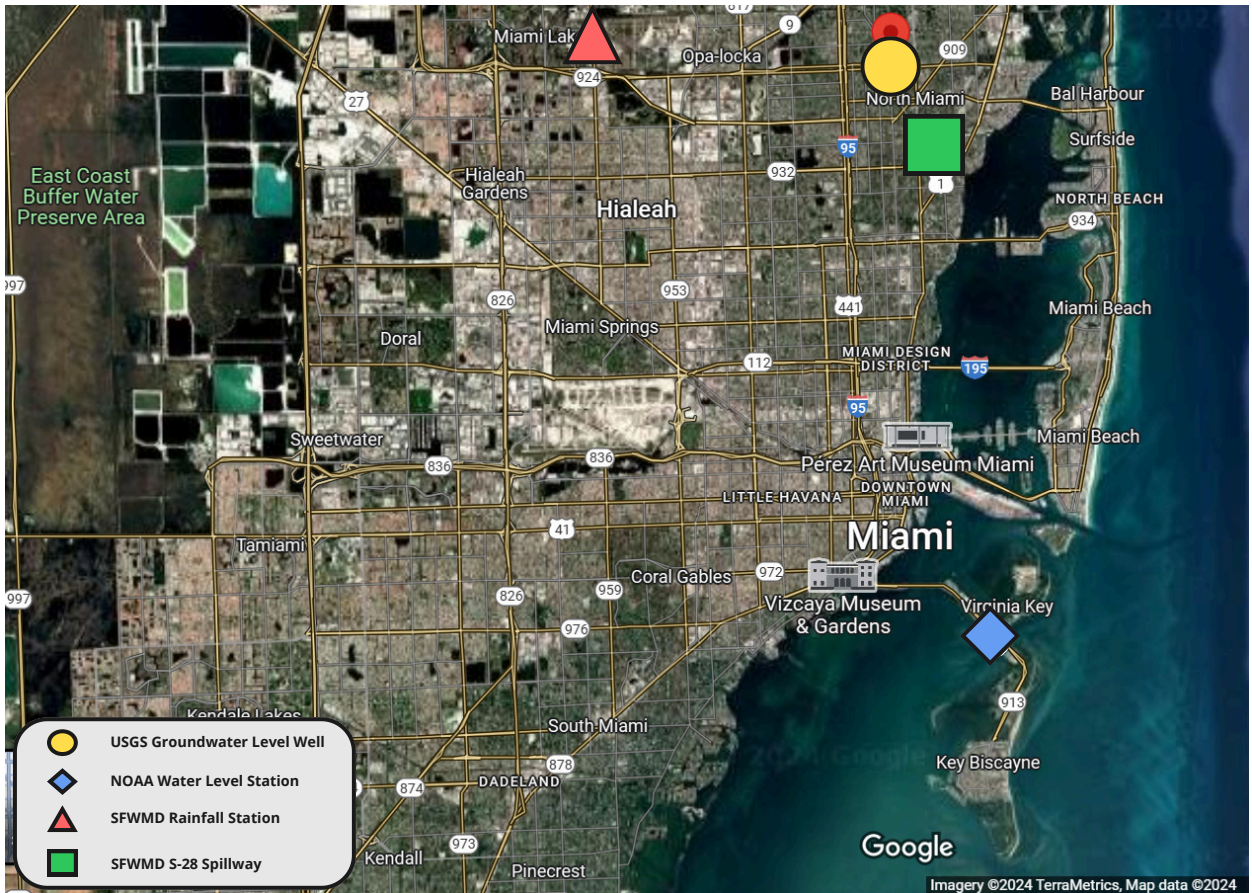


FIG. 1. Target location (gold circle) of USGS Groundwater Well, NOAA Tides & Currents Water Level Station 8723214 Virginia Key, Florida (blue diamond), South Florida Water Management District (SFWMD) Rainfall Station (red triangle), and the SFWMD S-28 Spillway (green square).

and requiring less compute power than physical models, with the trade-off that the spatial coverage of the models is usually much smaller. The next few paragraphs will discuss some examples of these existing methods.

Currently, physical models exist that are capable of simulating flooding events. Few of these studies focus on groundwater level as the main indicator of flooding. Sukop et al. (2018) created a 1-D groundwater flow model for the Arch Creek Basin in Miami-Dade County using the USGS MODFLOW model. Peña et al. (2022) created a modeling framework using FLO-2D and MODFLOW-2005 that simulates surface–subsurface water interactions and used it to analyze the relevance of groundwater flooding in the same area of Arch Creek Basin. However, physical models

take a considerable amount of time and resources to run, and make predictions or simulations for multiple grid points in an area.

Data-driven methods, such as the proposed ML solution in this work, tend to serve as a more flexible option for modeling since, once trained, ML models usually require less inputs than physical models. This is provided that there is historical data available that covers the different forcings for the event and the variable that is being predicted, and, if predicting a certain type of event, that there is a sufficient number of such events in the dataset. If the ML model inputs are available in real-time, then making the model operational is also, usually, an easier feat to accomplish than making a physical model operational.

Few scholarly publications have been published on the use of ML to predict compound flooding events. Moradian et al. (2024) paired the advanced multi-scale nested (MSN_Flood) hydrodynamic flood model and ML for short-term compound ocean-fluvial flood forecasting in Cork City, Ireland. Xu et al. (2025) used a 1D CNN to predict the spatial distribution of compound flooding in Haidian Island, Hainan Province, China. Shi et al. (2025) proposes Forecast Informed Deep Learning Architecture, FIDLAR, to achieve rapid and near-optimal flood management with precise water prereleases in a flood-prone coastal area in South Florida. Groundwater level was not used as input to FIDLAR. This work aims to further explore the ML approach to tackle the compound flooding problem at a location where such events have a large societal impact.

3. Data

The model features used as input to the ML models, as well as the target feature, groundwater level, are described below. For each input feature, the data source, the date range, and other useful metadata are described.

a. Groundwater level

Approved groundwater levels in feet above NAVD 88 were downloaded from the U.S. Geological Survey (USGS) water data API for the groundwater well G -3601S - 255358080114103.¹ This groundwater well is located in Miami-Dade County and sits atop the Biscayne Limestone Aquifer. It has a well depth of 75 feet. Data was downloaded for the years 2017 to 2024. The time series of groundwater level is illustrated in Fig. 2.

¹<https://api.waterdata.usgs.gov/>

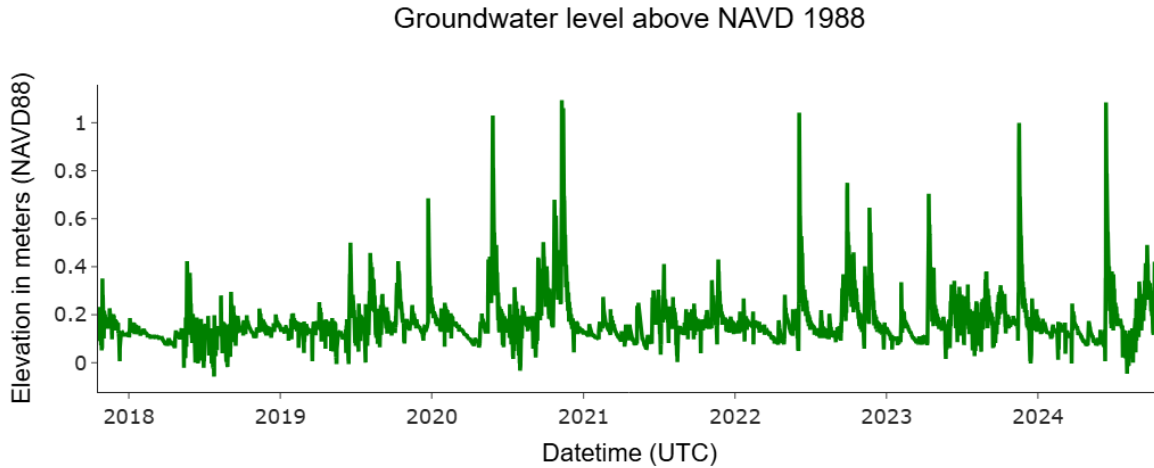


FIG. 2. U.S. Geological Survey (USGS) station G -3601S - 255358080114103 groundwater level time series.

b. Ocean Water level

Verified water levels in meters above NAVD88 were downloaded from NOAA Tides & Currents data API for Station 8723214 located in Virginia Key, Florida.² The time series was downloaded for the years 1998 to 2024. Only values from the same date range as the target groundwater levels were used for the ML models (2017-2024).

c. Rainfall

15-minute rain gauge data augmented with radar-based rainfall data was downloaded from the South Florida Water Management District (SFWMD) corporate environmental database DBHYDRO using their web-based application DBHYDRO Insights.³ The data was downloaded for station S28Z_R which is located in watershed C-8 (the same watershed as the target groundwater well location). The date range for the downloaded data is 2017 to 2024.

d. Spillway Parameters

Several parameters were downloaded for the S-28 spillway structure on canal C-8 at tidewater also using DBHYDRO Insights. These parameters include: telemetry (radio network) instantaneous gate opening data in feet for Gate 1 and Gate 2 at station S28_S, and telemetry instantaneous water

²<https://api.tidesandcurrents.noaa.gov/api/prod/>

³<https://www.sfwmd.gov/science-data/dbhydro>

level in feet above NAVD88 for station S28_H (Headwater) and station S28_T (Tailwater). The date range for the downloaded data is 2017 to 2024.

4. Model Design

This section will describe the different methods and techniques that were applied when building the ML models. Various methods were applied that aimed to generate robust and generalizable MLP models such as hyperparameter tuning, regularization techniques, k-fold cross-validation, and ensemble methods. The random forest (RF) baseline model is also briefly described.

a. Multilayer Perceptron (MLP)

The ML algorithm implemented in this work is the MLP. The MLP is a feed-forward neural network that uses activation functions, generally they are non-linear, in its hidden layer(s) to model the non-linear relationships between the model inputs and outputs and minimize the loss function (Hsieh 2023). MLP models are one of the most popular models for predicting groundwater level (Tao et al. 2022). The MLP models were trained using the Keras deep learning API in Python. The inputs and hyperparameters differ depending on the model. The different models are discussed in section 5. Each model used a ReLU activation function in the hidden layers and a sigmoid activation function in the output layer. An additional layer was added after the sigmoid output layer to denormalize the sigmoid output from the [0,1] range back to the original groundwater level scale. This denormalization was performed using the inverse of the min-max scaling equation:

$$y_{\text{denormalized}} = y_{\text{scaled}} \times (x_{\text{max}} - x_{\text{min}}) + x_{\text{min}} \quad (1)$$

The maximum groundwater level value (x_{max}) was increased by 20% to allow for possible extrapolation. The Adam optimizer algorithm, a stochastic gradient descent method, was used during training (Kingma and Ba 2017). A mini-batch size of 64 was utilized. 64 is a common mini-batch size chosen based on modern CPU and GPU requirements. Using mini-batches in stochastic gradient descent allows for better computational efficiency and lower memory usage. It is also less likely to converge to shallow local minima compared to the traditional batch training method which uses the full dataset during gradient descent (Hsieh 2023).

TABLE 1. Grid search space for hyperparameter tuning.

Hyperparameter	Values
Hidden Neurons	50, 100, 200
Hidden Layers	1, 2, 3
Learning Rate	0.001, 0.01, 0.1
Batch Size	16, 32, 64

b. Random Forest (RF)

RF models were also trained and used as a baseline model to compare to the more complex MLP models. RFs are an ensemble of decision trees where each tree is trained on a subset of the training data to reduce variance and avoid overfitting; randomness is introduced using techniques such as bagging and random feature selection at each split (Breiman 2001). Previous studies show that RFs have been used to model groundwater level, ocean water level, and flood risk. (Tang et al. 2019; Ayus et al. 2023; Sadler et al. 2018; Muñoz et al. 2024). The RF models were tuned by comparing performance while varying the number of trees (100, 200, 1000), the maximum depth of the trees (None, 10, 20), the minimum number of samples required to split a tree node (2, 5), and the maximum number of features to consider when looking for the best split (sqrt, $\log_2$). Most of the hyperparameters of the selected RF models were in the middle of the investigated ranges with 100 trees, a maximum depth of 10, max logarithmic features and a split of min samples of 2 or 5, indicating that more complex RF models would have been unlikely to improve performance, including during extreme precipitation events. The performance of the MLP models is compared with the baseline RF (as well as linear regression) in section 6.

c. Hyperparameter Tuning

Hyperparameter tuning was implemented during model training and the performance of each model documented such that a model could be selected which performed well based on the performance metrics in section 5. The hyperparameters that were tuned are shown in Table 1. High numbers of hidden neurons were used due to the high complexity of the model. Each model that was trained consisted of at least 175 input nodes each. More information on how these input time series are handled can be found in section 5. A shallow neural-network (3 or fewer

hidden layers) was chosen after testing for larger models and selecting the least complex model achieving similar high performance. Model performance was most sensitive to the initial learning rate controlling the size of the initial step during gradient descent. The grid search algorithm was used for tuning. Mean squared error (MSE) was used as the loss function. The validation loss was used as the objective function to choose the optimal hyperparameters.

d. Regularization

To avoid overfitting, early stopping (Morgan and Bourlard 1989) and dropout (Srivastava et al. 2014) were implemented. The performance of the models on both the validation and testing sets was used as evidence of the generalizability of the model. As part of the dropout process, input nodes typically have a smaller probability of being deleted (around 20%) while hidden nodes are more likely to be deleted during training (around 50%). This dropout technique allows for the creation of multiple model architectures during a single training session (Hsieh 2023).

e. K-fold Cross-validation

K-fold cross-validation was implemented during hyperparameter tuning to account for the variability from year to year in the dataset (Yadav and Shukla 2016). The years 2017-2023 were used for the cross-validation, where each year was used exactly once as a validation set. 2024 was left out of the training process such that it could be used as an independent test set. This resulted in seven folds and thus seven rotations total. During hyperparameter tuning, to choose the best model, the performance of each hyperparameter combination from the GridSearch was compared across all seven rotations. The hyperparameters that produced models which performed the best on average across all folds (based on mean validation loss) were chosen as the optimal hyperparameters to use for the final ensemble members. The ensemble method is discussed further in Section 4f.

f. Model Ensemble

To account for the stochastic nature of the MLP model, an ensemble approach was taken. In this case, 30 ensemble members were trained using the optimal hyperparameters found from the k-fold cross-validation hyperparameter tuning process described previously. Each member was trained using a different random seed to introduce weight initialization diversity. Then, the median of the

TABLE 2. The name of each model and the description of the different input features used by each model.

Model	Description
Model 1	Uses only past measurements as inputs.
Model 2	Uses past measurements and ocean water level predictions as inputs.
Model 3	Uses past measurements and rainfall predictions as inputs.
Model 4	Uses past measurements and gate 1 and 2 opening predictions as inputs.
Model 5	Uses past measurements and ocean water level, rainfall, and gate 1 and 2 opening predictions as inputs.

ensemble member predictions was used for the forecast rather than just a single model run (Hsieh 2023). This helped to reduce the variance error that comes with only running a single model.

5. Experimental Design

This section discusses the models selected to determine which time series predictions or combination thereof was useful for predicting the groundwater level at the target location. This experiment also explored whether the relatively small number of input features is sufficient for accurate predictions at the location, further justifying the low complexity approach of the MLP. The different performance metrics that were used to evaluate the models are also described.

An experiment was carried out that investigated how using time series predictions for certain input variables affects model performance. Time series predictions were created using perfect prognosis, a method that treats observations as predictions when creating samples (Wahl et al. 2015). Perfect prognosis assumes that perfect predictions are available at inference time. The experiment involved training multiple MLP models where each model utilizes different input variables. Some models do not use time series predictions for any input variable. Some models use time series predictions for only one input variable. Lastly, some models use time series predictions for multiple input variables. This was done for each lead time (3, 6, 12, and 24 hours). Each model is described in detail below.

- 1. Measurements Only:** model 1 utilized only past measurements to predict the target for each lead time, no predictions. This model was used as a baseline model to compare it with the other models. The following models added predictions for various features as inputs.

- 239 **2. Predicted Ocean Water Levels:** model 2 utilized ocean water level predictions and past
240 measurements for each feature to predict the target at each lead time. This model showed
241 whether or not ocean water level predictions led to improved performance when compared to
242 model 1. Such predictions would come from NOAA tidal predictions or more sophisticated
243 models such as operational ML model predictions (Estrada et al. 2025).
- 244 **3. Predicted Rainfall:** model 3 utilized rainfall predictions and past measurements for each
245 feature to predict the target at each lead time. This model showed whether or not rainfall
246 predictions led to improved performance over model 1. Such predictions could come from
247 the High Resolution Rapid Refresh (HRRR) model for the area.
- 248 **4. Predicted Gate Opening:** model 4 utilized gate 1 and 2 opening predictions as well as
249 measurements for each feature as inputs to predict the target at each lead time. This model
250 showed whether or not gate opening predictions improved performance when compared to
251 model 1. This model served as a scenario in which future gate opening data was made
252 available ahead of time during the time of prediction such as when testing response scenarios.
- 253 **5. Predictions All Inputs:** model 5 utilized perfect prognosis predictions for ocean water levels,
254 rainfall, and gate 1 and 2 opening. Past measurements for all features were also used as inputs.
255 This model gauged model performance after adding predictions from multiple features as
256 inputs.

259 To generate samples for the input features and the target variable for training, the sliding window
260 technique was implemented (Shi et al. 2025). Fig. 3 visualizes this technique. Each model that was
261 trained used the past 24 hours and the current (lag hour zero) measurements for each variable as
262 inputs. Some models utilize predictions for certain input variables (such as rainfall). The number
263 of predictions used for a certain input variable depends on the lead time. Headwater and tailwater
264 predictions were not used as inputs to any model due to the fact that predictions for these variables
265 in real-time would be unrealistic. Groundwater level predictions were not used as inputs, as it is
266 the target variable and would provide an unfair advantage to the models.

267 To further gauge performance, each model was tested on the independent test set of 2024. The
268 performance of each model during confirmed flooding events was also analyzed. This flood event
269 case study is further described in section 6. Various metrics were used to gauge the performance

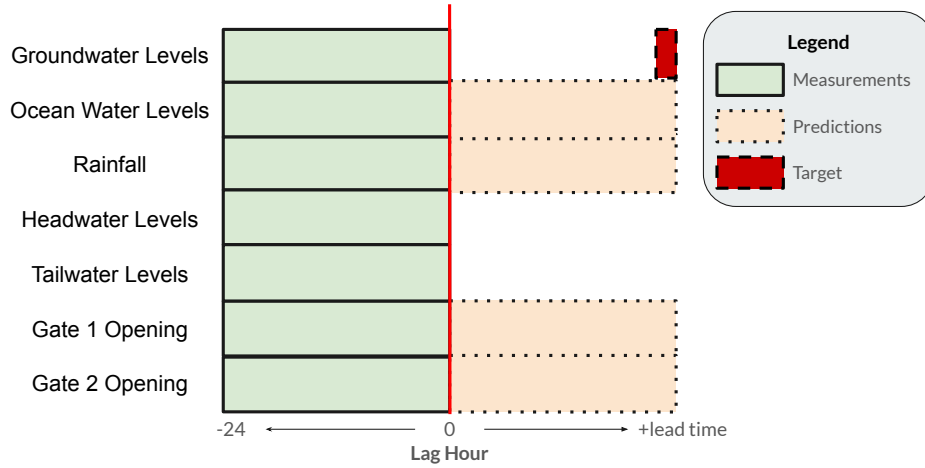


FIG. 3. Sliding window technique used to generate training samples for the input features and the target variable (groundwater level).

of the models. One metric being central frequency. Central frequency calculates the percentage of predictions within N cm of observations. For this study, the central frequency within 15 (CF15CM), 10 (CF10CM), and 5 cm (CF5CM) were used as metrics. Other metrics used are mean squared error (MSE), root mean squared error (RMSE), mean absolute error (MAE), and R^2 , which are some of the most widely used performance metrics for ML models predicting groundwater level (Tao et al. 2022).

6. Results

In this section, the performance of each model is assessed through tables and plots comparing the respective metrics on the full independent test set. The performance of each model is compared across all lead times with the performance of the most accurate model also compared to the baseline linear regression (LR) and random forest (RF) models.

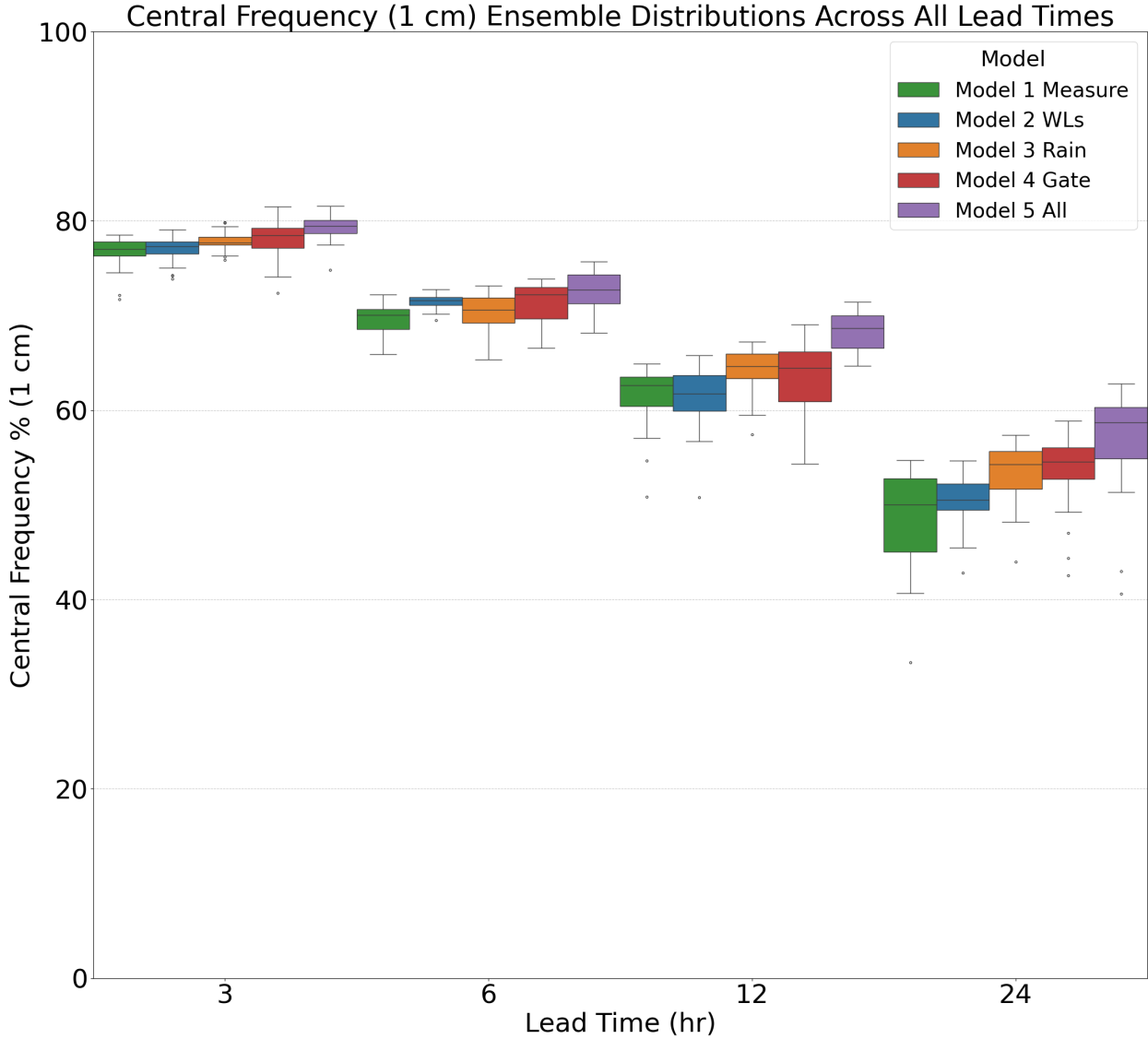
Table 3 shows the performance of the different models for growing lead times with performance decreasing, as expected, for longer lead times. While the addition of predictors results in better performance, the improvements vary substantially. Adding predicted ocean water levels leads to minimal improvements. Providing the model with predicted rainfall improves performance more significantly for all lead times and metrics. The biggest improvements are not surprising for the

TABLE 3. Model performance across lead times. Each row shows the performance for a different model at all lead times for each performance metric.

Model	Lead Time (h)	CF (15 cm)	CF (5 cm)	CF (1 cm)	RMSE (cm)	R ²
Measurements Only	3	99.8	98.5	79.3	1.8	0.98
	6	99.7	96.7	72.4	2.6	0.95
	12	99.5	94.6	64.3	3.6	0.90
	24	98.9	89.7	53.2	5.0	0.82
Predicted Ocean Water Levels	3	99.8	98.5	79.6	1.7	0.98
	6	99.7	96.8	72.4	2.6	0.95
	12	99.5	94.9	65.3	3.6	0.90
	24	99.0	90.4	53.3	4.9	0.82
Predicted Rainfall	3	99.9	98.5	80.4	1.5	0.98
	6	99.8	96.8	73.6	2.2	0.97
	12	99.7	94.7	67.3	2.7	0.94
	24	99.5	90.3	57.0	3.5	0.91
Predicted Gate Opening	3	99.8	98.7	81.3	1.6	0.98
	6	99.7	97.4	75.0	2.3	0.96
	12	99.5	95.4	68.3	2.7	0.94
	24	99.1	91.2	57.1	3.4	0.92
Predictions All Inputs	3	99.9	98.7	82.3	1.5	0.98
	6	99.7	97.4	75.5	1.9	0.97
	12	99.7	96.4	71.7	2.2	0.96
	24	99.8	93.6	62.3	2.7	0.94

longest lead time since perfect prognosis values are provided. Correlation improves from 0.82 for models 1 & 2 to 0.91 for model 3 while CF(1 cm) improves from 53.2% and 53.4% respectively to 57.0% all for 24-hour lead time predictions. Adding the predicted gate openings as a predictor also results in significant improvements, relatively similar to adding predicted rainfall predictions to the model. Comparing performance for 24-hour lead time, the model with predicted rainfall has a higher CF(15cm) at 99.5% vs 99.1% for the gate openings while for the CF(1cm) metric the performance with the gate opening predictions is minimally better than with predicted rainfall, 57.1% vs 57.0%. For all lead times, the model "Predictions All Inputs" produced equal or better performance metrics as compared to all the other models with the largest performance gain obtained for the 24-hour predictions and CF(1cm) at 62.3% with all inputs as compared to the other models having a performance of 57.1% or lower for the same metric. These same trends can be seen in Fig.

299 4 which shows a box plot of the central frequency within 1 cm (CF1CM) ensemble distributions
 300 across all lead times.



301 FIG. 4. Box plot of the central frequency within 1 cm (CF1CM) ensemble distributions for each model across
 302 all lead times.

303 Comparing the best model, "Predictions All Inputs", across all lead times, performance increases
 304 as lead time decreases. This is most notable in the CF(1 cm) metric, where performance improves
 305 from 62.3% for the 24-hour lead time model to 82.3% in the 3-hour lead time model. Fig. 4
 306 shows the same trend when comparing the CF(1 cm) performance across lead times while also
 307 displaying the performance variability across the 30 trained ensemble members for each model

and lead time. While there is overlap in the respective performance ranges, performance decreases with lead time from just below 80% for 3 hour lead times to CF(1 cm) in the 50% range for 24 hour lead times. And for each lead time, there is a trend of increasing performance from model 1 with only measurements as predictors through model 5 with all the inputs included.

TABLE 4. Model 5 ("Predictions All Inputs") performance compared to baseline models for all lead times. Each row shows the performance of each model architecture at a different lead time for each performance metric.

Lead Time (h)	Model	CF (15 cm)	CF (5 cm)	CF (1 cm)	RMSE (cm)	R ²
3	LR	99.9	99.1	84.0	1.4	0.99
	RF	99.7	96.4	72.3	2.6	0.95
	MLP	99.9	98.7	82.3	1.5	0.98
6	LR	99.8	97.8	75.2	2.1	0.96
	RF	99.6	95.2	68.6	3.1	0.93
	MLP	99.7	97.4	75.4	1.9	0.97
12	LR	99.6	95.6	67.6	2.9	0.94
	RF	99.2	92.8	63.2	3.6	0.90
	MLP	99.7	96.4	71.7	2.3	0.96
24	LR	99.2	90.5	54.7	3.9	0.89
	RF	98.8	89.3	52.9	4.3	0.87
	MLP	99.8	93.6	62.3	2.7	0.94

Lastly, Table 4 compares the performance of the MLP model with the baseline LR and RF models for all lead times. All the MLP models used for this comparison are with all inputs included in the "Predictions All Inputs" model. At longer lead times (12 and 24 hours), the MLP model outperformed the baseline models for all metrics. However, as the lead time got smaller (3 and 6 hours), the LR model performed similarly to the MLP model, even surpassing the MLP model at the 3 hour lead time. The RF models equaled or mostly underperformed the LR and MLP models performance for all lead times.

7. Discussion

This section discusses and compares the respective improvements brought by the different predictors versus a model based only on past measurements. The performance of the models during

confirmed flooding events is also analyzed. The importance of rainfall predictions accuracy is investigated. Lastly, the MLP model is compared to the simpler baseline models, LR and RF, during an extreme event.

a. Advantages of time series predictions as input

When considering all metrics presented in Table 3, there is no significant advantage to adding predicted ocean water levels as input to the "Measurements Only" model or model 1. Small improvements in the spread of the confidence intervals can be seen in Fig. 4, for the model 2 24 hour predictions as well as for the 6 hour predictions for which model 2 has the smallest confidence interval of all models. This suggests that ocean water level predictions could have a small contribution to reducing uncertainty in the predictions, but this would require further statistical analysis for all lead times. In general, these results suggest that ocean water level predictions were not useful to significantly improve groundwater level predictions at this target location. This could seem surprising for a coastal location and as the tidal influence is clearly visible in the groundwater level time series as illustrated in Figures 5 and 7. It is hypothesized that ocean water level predictions are not helpful because (1) large changes in groundwater levels for this data set and location are not correlated with significant water level surges in the nearby bay and that all coastal inundation events within the study period were driven by precipitation and (2) given that tidal fluctuations are highly predictable, the ML models can likely predict future tidal influence based on recent measurements, which include a daily tidal cycle.

The models that utilized rain or gate opening predictions as input, models 3 and 4, showed an improvement in performance over models 1 and 2. This was most noticeable for CF(1 cm), especially for the longer lead time models (12 and 24 hours). This shows that having accurate rainfall or gate opening predictions as input provides information that significantly improves groundwater level predictions at the target location. There is however a significant difference between these two predictors. The opening of gates along the canal are human controlled and in part driven by the dynamic of flooding events. Hence, gate openings are in part driven by precipitation. In terms of modeling, and eventual research to operation transition, the accuracy of precipitation predictions is the most important factor based on these results for this location. Section 7b analyzes model predictions during confirmed flooding events and will help further analyze how rainfall and

gate opening predictions affect groundwater level predictions, while section 7c investigates how precipitation forecast accuracy influences the performance of the models.

Model 5, the model that uses all the available predictors, ocean water level, rainfall, and gate opening predictions, performed best overall based on its performance over the independent test set. This performance is likely driven by the combination of rainfall and gate predictions being added as input given the individual improvements provided by these two predictors demonstrated for models 3 and 4.

Although gate opening predictions improve the forecast of groundwater levels, in an operational setting this model would require a planned schedule for the gates, which is unrealistic as gate openings take into account the state of the canal and the overall flooding situation in the study area which both depend in large part on evolving precipitation. However, having a model with gate openings as a predictor could still be useful to compare scenarios as flooding events unfold, although the relationship between precipitation and gate openings should be further studied. More research would be needed to explore this potential advantage to determine whether the use of gate predictions in real time is viable or not. The model including gate predictions could also be used to test different flood scenarios and help flood management planners explore how adjusting gate operation during these extreme events affects groundwater levels.

b. Case study: extreme event visualization

Several flooding events in the study area can be identified in the groundwater level time series and confirmed based on local records such as press records. Fig. 5 shows four cases in which there were large spikes in groundwater level at the same time as recorded flooding events in Miami-Dade County, Florida (Harris 2023). Events (a), (b), and (c) are part of the training set, while event (d), which took place in June 2024, is part of the independent test set.

Fig. 6 shows the performance of each 24-hour MLP model during the June 12, 2024 flood event. The observed groundwater levels are the solid black line. Each model prediction time series is the median of the predictions from all 30 ensemble members for that model.

The prediction time series line of model 2 is almost identical to that of model 1. This aligns with the points discussed in Section 7a, suggesting that utilizing ocean water level predictions as input for predicting groundwater level at the target location is not useful. While water levels could have

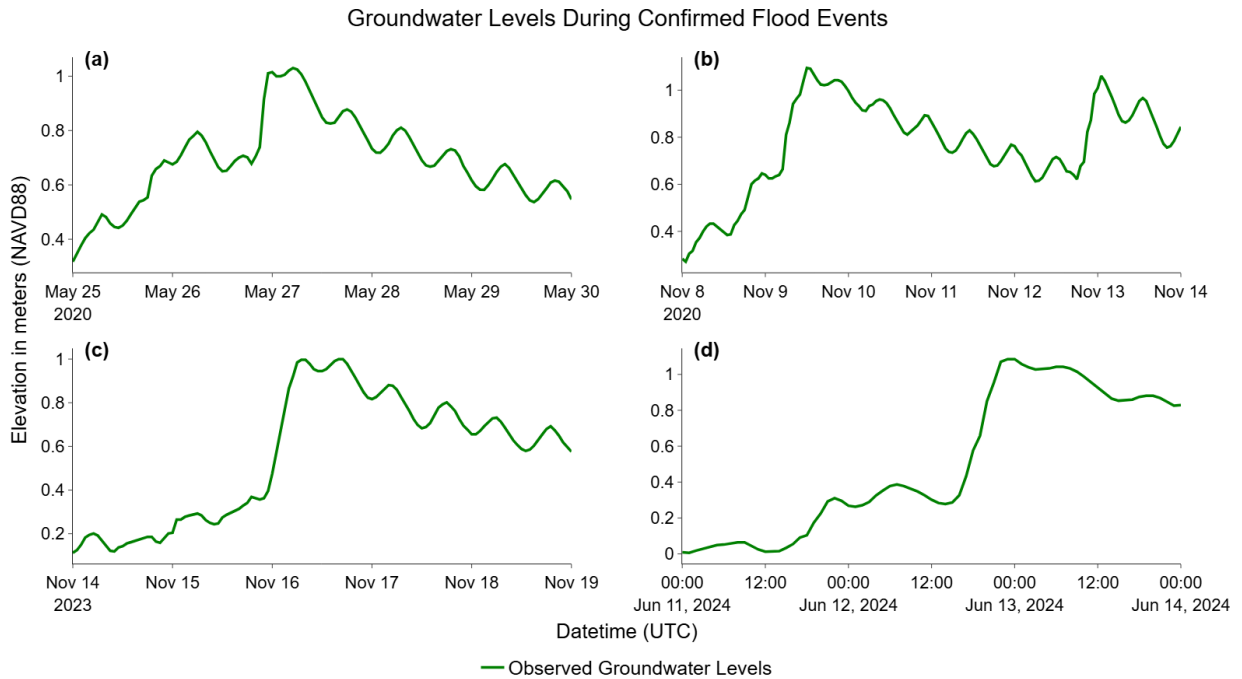


FIG. 5. Groundwater level conditions during 4 different flooding events in Miami-Dade County, Florida. Each subplot represents a different flooding event.

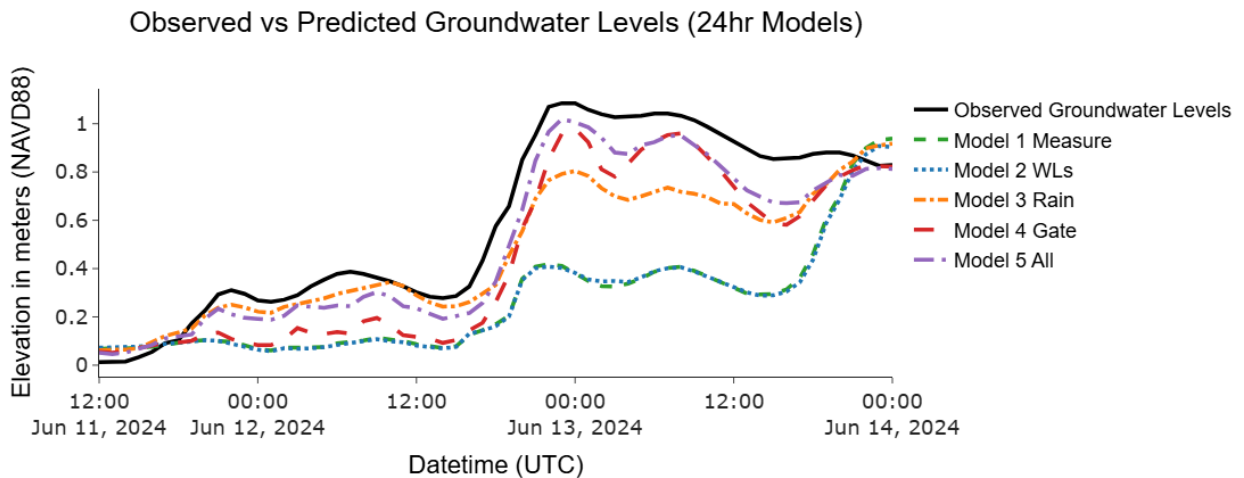


FIG. 6. Groundwater level observations and predictions for all 24-hour models during the June 12, 2024 flood event. Each model prediction time series is the ensemble median of all 30 ensemble members for that model.

influenced the dynamic of a flooding event, the results show that such predictions are not helpful during this study's flooding events likely because these events are all driven by precipitation and

388 there are no rapid water level changes which predictions could have benefited performance. If a
389 flooding event were to be driven in large part by bay water levels, these models predictions would
390 not be appropriate, as such events are not part of the training or overall study data.

391 Model 3, the model using rain predictions, was able to capture the ramp up to the event relatively
392 well, while also having smaller prediction errors during the peak of the flood compared to models
393 1 and 2. Model 4, the model that used gate predictions but not rain predictions, struggled at the
394 beginning of the event but was able to predict the flood peak relatively well. This shows that model
395 3 is able to use rain predictions to predict the gradual increase in groundwater level due to rain, but
396 may be missing some information that allows it to more accurately predict the peak of the flood.
397 Model 4 predicts the peak of the flood well, but does not have enough information to predict the
398 start of the event on its own.

399 Model 5 acts as a combination of models 3 and 4 in the way that it was able to predict the whole
400 flood event the best compared to all other models. This shows that utilizing rain and gate opening
401 predictions in tandem is useful for predicting groundwater level during flooding events. However,
402 as mentioned in Section 7a, using gate opening predictions as input in real-time would be difficult,
403 although operators could use the model as the flooding events unfold to compare the timing of
404 gate openings. The overall performance of model 2 suggests that ocean water levels do not aid the
405 performance of model 5 as much as the rain and gate opening predictions and can likely be omitted
406 from the model.

410 Fig. 7 shows the performance of the "Predictions All Inputs" model for each lead time during
411 each flood event shown in Fig. 5. For the most part, shorter lead time models (3 and 6 hours) were
412 able to predict groundwater level with low error during these flood events. However, there are some
413 periods during the flood events where the longer lead time models (12 and 24 hours) produced
414 better predictions. This is most notable during events (c) and (d), where the 24-hour model was
415 able to capture the peak of the flood slightly better than the shorter lead time models. It is expected
416 that a model that predicts a flood event further in the future would struggle more than a model that
417 is only predicting 3 hours in the future. The reason for this anomaly is not entirely known, but
418 it may be that based on the inputs provided, given that the 3 and 6 hour models only use 3 and
419 6 hours of predictions for each input, respectively, that there is not a large enough accumulation
420 of information in the short lead time before a flood event that allows the model to predict the full

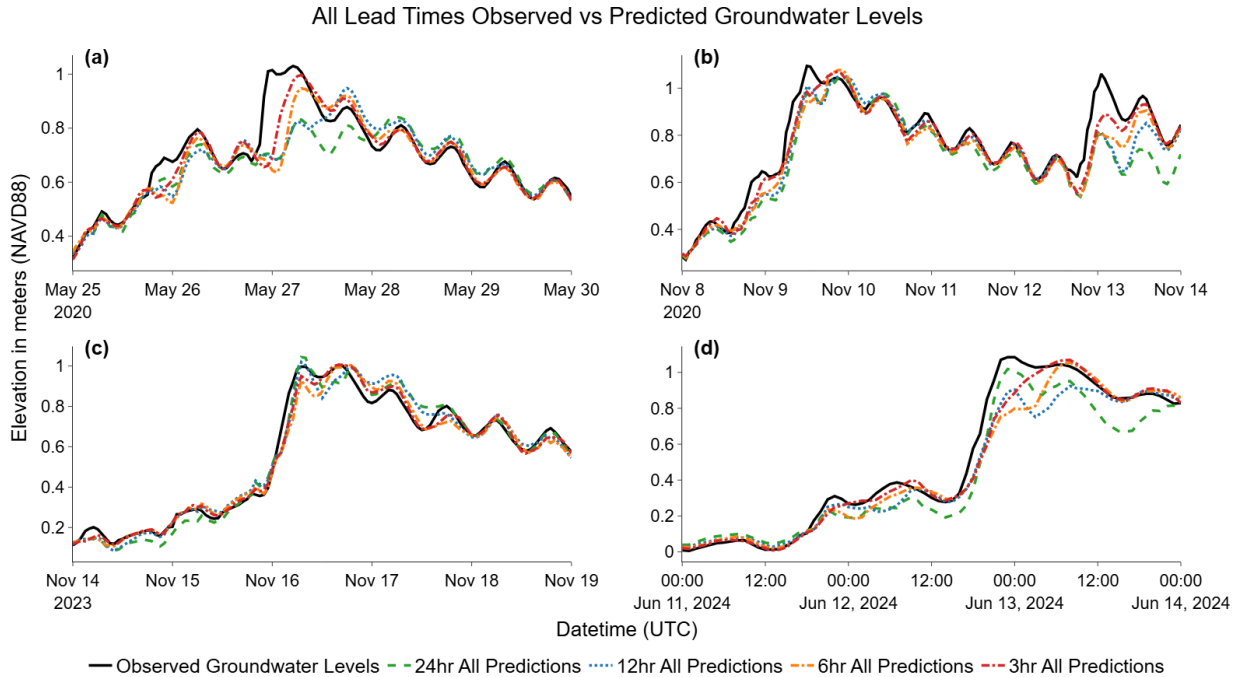


FIG. 7. Groundwater level observations and predictions for the "Predictions All Inputs" model at each lead time during confirmed flooding events. Each model prediction time series is the ensemble median of all 30 ensemble members for that model.

magnitude of the flood event on time. More work would need to be done to investigate the reason for this disparity.

c. Sensitivity to rainfall predictions accuracy

This section investigates the sensitivity of the groundwater predictions to the accuracy of the rainfall predictions during confirmed flooding events. For this analysis, we use model 3 which only uses rainfall predictions and past measurements as input. Specifically, we ran model 3, with a 24 hour lead time, multiple times. Each time, the rainfall predictions were scaled by a different value ranging from 0.00 to 3.00 with increments of 0.05. A scale factor of 0.0 means that no rain was predicted, 1.0 means that the rainfall predictions are not altered in any way, i.e. perfect prognosis of precipitation, and 3.0 means that the rainfall prediction totals were tripled and thus more intense. Note that for each case, the rainfall measurements were not altered in any way, only the rainfall predictions. For each rainfall prediction scenario, the ensemble median of model 3 was recorded.

Flood-event Sensitivity to Rainfall Prediction Intensity Scaling

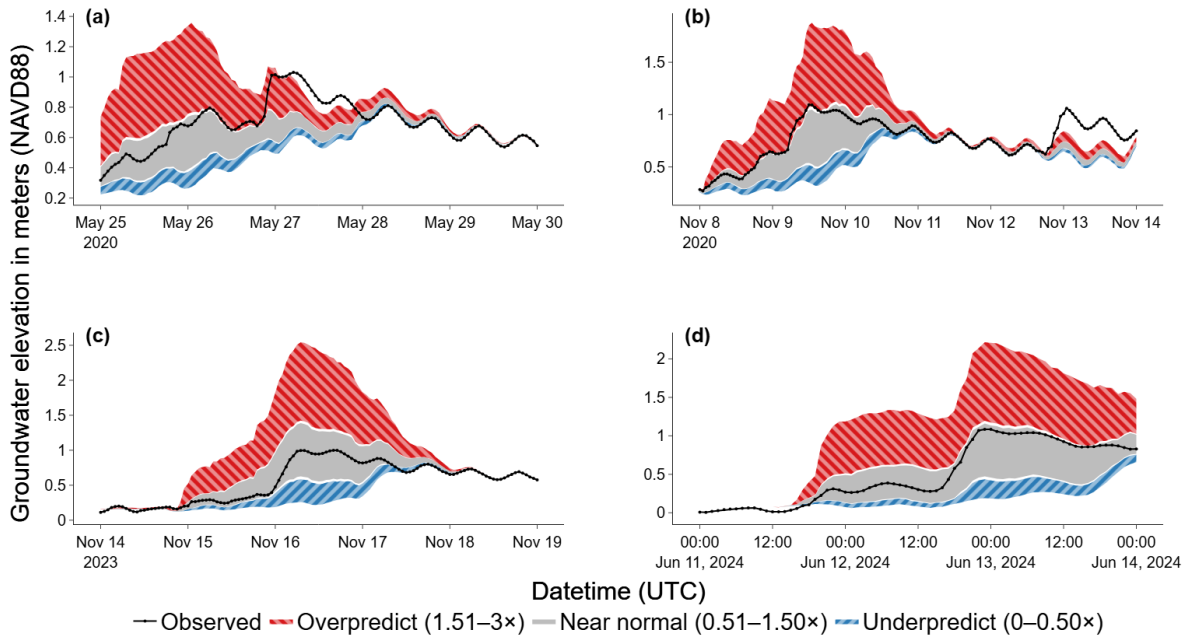


FIG. 8. Model predictions are from the 24 hour "Predicted Rainfall" Model. Each subplot shows how the model groundwater level predictions compared to observations (black line) during confirmed flooding events when the intensity of the rainfall predictions used as input to the model is scaled. Each ribbon shows the model predictions when the rainfall predictions are scaled by a certain scale factor (Underpredict, Near normal, and Overpredict). For each rainfall prediction intensity scenario, the ensemble median is used as the model prediction.

Fig. 8 shows the sensitivity of the model to rainfall 24-hour predictions during the four confirmed flooding events mentioned throughout this study. For the analysis and the comparative visualization, the scaled rainfall predictions are regrouped using three scenarios: when the rainfall totals were over-predicted (1.51x–3.0x), near normal (0.51x–1.50x), or under-predicted (0x–0.50x). The ribbon plots show that whenever the future rainfall totals are under-predicted, the model struggled to predict the groundwater level during a flood. And whenever the rainfall predictions are at or near normal, the model is able to accurately predict during most of the flooding events, though not all.

It also appears that a higher predicted rainfall intensity (closer to 1.5–3 times the normal predicted amounts) led to more accurate predictions at some points in the floods, particularly for the secondary increase in groundwater after the main large increase in groundwater levels in event (b). Considering that the "rainfall predictions" in this case are actually observed rainfall totals treated as predictions

(perfect prognosis), this may suggest that either the source of rainfall measurements used in this study does not always accurately represent the rainfall conditions including the exact geographic distribution of the rainfall, or that the ML model accuracy decreases when predicting the most extreme events. The latter can result from the sparsity of extreme events in the training data and could be improved upon by further improving the training of the models or adopting a hybrid modeling approach (Chen et al. 2024; Sham et al. 2025). The rainfall data is based on rain gauge data from a rainfall station that is a little more than 6 miles away from the target location. The rain gauge data is then augmented with radar-based rainfall data for the SFWMD watershed C-8. While this is a high quality data set, this could still be a source of error during precipitation events (Terink et al. 2018; Villarini et al. 2008; Li et al. 2023) and it may be worth further investigating whether the source of rainfall used in this study could be improved for both the intensity of the rainfall predictions and the location of the precipitation.

However, Fig. 8b shows that accurate rainfall may not be enough for accurately predicting groundwater level during these extreme events. Even after scaling the rainfall predictions to be 3 times more intense than they actually were, the model was still not able to accurately predict during the November 13, 2020 flood event, even though the model was able to predict the November 9, 2020 event only a few days earlier. It could be that the Nov. 13, 2020 event is different from the Nov. 9 event, and that the model is missing information needed to predict this event. Or it could be that since the floods are so close in time to each other, that the model struggled to predict the second event since the groundwater level was still recovering from the first event. It should be noted that these Nov. 2020 events occurred during Tropical Storm Eta, a storm that consisted of heavy rainfall and tropical-storm-force winds.

d. Baseline model comparison during an extreme event

A comparison of the MLP predictions with the simpler architectures tested for this study, LR and RF, is presented in Figure 9. Specifically, the absolute prediction errors (the absolute value of the difference between observation and prediction at each time step) are compared for the "Predictions All Inputs" predictors and a 24 hour lead time during the June 12, 2024 flood event. The MLP model shows smaller absolute errors throughout the entire flooding event. Both the RF and LR models are unable to predict the peak of the flooding event with peak absolute error around 0.5

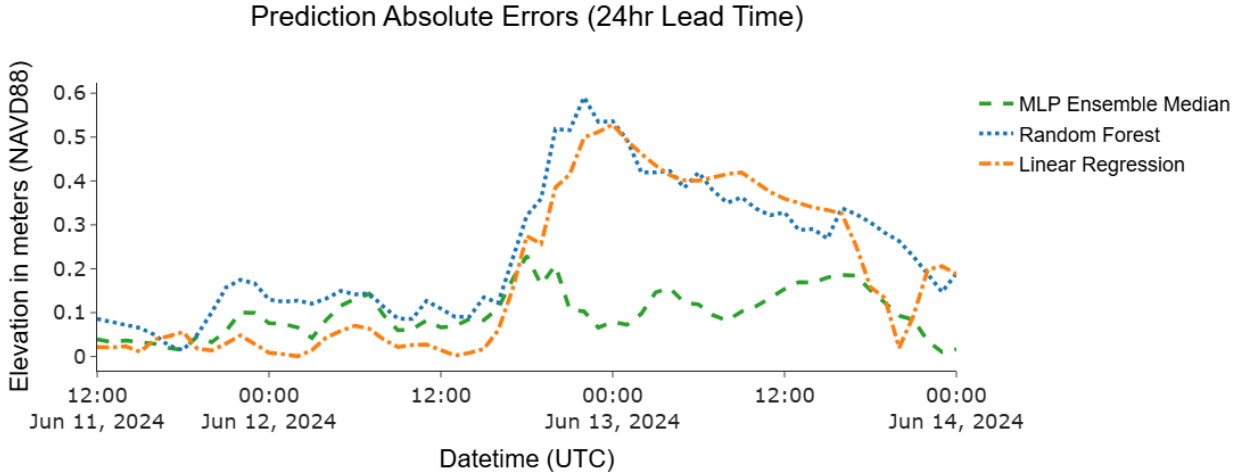


FIG. 9. Absolute prediction errors for the 24 hour "Predictions All Inputs" MLP model compared to the baseline linear regression (LR) and random forest (RF) models during the June 12, 2024 flood event.

m to 0.6 m during the period of June 12 and June 13 as compared to a peak error below 0.3 m, about 50% lower, for the MLP model. Table 4 provides a broader comparison for the full test year showing a lower RMSE and a higher CF(1 cm) for all metrics for lead times larger than 6 hours. These comparisons provide the justification for the use of the more complex MLP since the simpler models were unable to match its performance and capture the dynamic of the flooding event in the test set.

It should be noted that during the June 12, 2024 flood event, the 3 hour LR model performed similarly to the MLP model. However, during the flooding event, the LR model mostly displayed behavior that matches that of a persistence model. This suggests that the LR model simply uses the most recent groundwater level measurement as the prediction itself. The fact that the 3 hour MLP model was not able to outperform the LR model during the flood event suggests that more information or predictors could further improve performance.

8. Conclusion

Miami, Florida is highly susceptible to compound flooding due to its low elevation, high groundwater table, and proximity to the Biscayne aquifer. Flood dynamics in the region are influenced by precipitation, groundwater levels, and coastal conditions, while canal management used for salt-

496 water intrusion mitigation can further affect flood risk during extreme rainfall events. Improving
497 predictive modeling of these events is therefore critical for preparedness and mitigation.

498 This study developed machine learning models to predict groundwater level as a proxy for
499 compound flooding events in Miami. Using a 7-year dataset (October 2017–December 2024),
500 multilayer perceptron (MLP) models were trained to forecast groundwater levels at lead times of
501 3, 6, 12, and 24 hours. The year 2024 served as an independent test set for evaluating model
502 performance and analyzing predictions during confirmed flooding events.

503 An experiment investigated the impact of including prediction time series for certain inputs
504 compared to relying solely on past measurements. Multiple models were trained with different
505 combinations of predictors, including ocean water level, rainfall, and spillway gate opening predic-
506 tions. Model performance was also compared with simpler baseline approaches including linear
507 regression and random forest models.

508 Results showed that models incorporating predictions for rainfall and gate openings produced
509 the best performance across the test set and during flood events. These predictors provided
510 valuable information for capturing groundwater dynamics, particularly for longer lead times. In
511 contrast, ocean water level predictions provided minimal improvement. The MLP models generally
512 outperformed the baseline models at longer lead times (12–24 hours), supporting the use of a neural
513 network architecture for this application.

514 The models were able to predict groundwater levels during confirmed flooding events with
515 relatively low error, even for longer lead times. However, shorter lead-time models sometimes
516 performed similarly to linear regression during extreme events, suggesting that additional predictors
517 or further model refinement may improve performance.

518 A sensitivity analysis further demonstrated that rainfall prediction accuracy strongly influences
519 groundwater forecasts. When rainfall totals were underpredicted, model performance degraded
520 substantially during flood events. Near-accurate rainfall predictions produced the most reliable
521 groundwater forecasts, emphasizing the importance of high-quality precipitation forecasts for
522 operational use.

523 Overall, the results indicate that machine learning models using groundwater level as a proxy can
524 effectively predict compound flooding events at the target location in Miami-Dade County, Florida.
525 Accurate rainfall predictions remain a key requirement for operational implementation, while the

526 use of gate operation predictions requires further investigation to determine their practicality in
527 real-time forecasting systems.

528 **9. Future Work**

529 The models performed well with the use of perfect prognosis predictions. However, attempting to
530 use the models to predict groundwater level in real-time requires an accurate source of predictions
531 for each feature to be used in place of these perfect predictions. Adding rainfall predictions as
532 an input to the models resulted in a substantial increase in performance. Thus, finding and/or
533 improving the most real-time rainfall predictions to use as inputs for the model would be the next
534 step towards achieving this goal.

535 Using gate opening perfect predictions as inputs to the models also led to an increase in model
536 performance. However, gate opening data are dependent on control schedules that are based on
537 rule-based methods which decide when and how much to open the gates based on observed data.
538 Research has already been done towards optimizing these control schedules for flood mitigation
539 using ML methods (Shi et al. 2025). A similar system could be implemented to generate control
540 schedules for the gates used in this study to optimize how the gate should operate in real-time
541 during a flooding event. These predictions could then be used as inputs to the models.

542 There also exist pumps, another hydraulic structure that is used to control the flow of water
543 during flooding events. Pump data was not used as an input to the models for this study. Future
544 work could be done towards adding pump data to the models and observing how it affects model
545 performance. However, even without pump data, this study showed that the models were able to
546 accurately predict the groundwater level at the target location.

547 Lastly, there were correlations between the groundwater level time series used in this study and
548 flooding events in the target location of Miami-Dade County, Florida. Still, more research could
549 be done to explore this relationship. For example, a sensor could be installed near the target
550 groundwater well that measures surface water during flood events. This could provide a better
551 indicator of how high the groundwater level needs to rise before the surrounding surface area
552 begins to become inundated.

Acknowledgments. This material is based upon work supported by the National Science Foundation under awards 2019758 and 2331908. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation.

Data availability statement. Publicly available data was used for this study. The code to process the data and train the ML models can be found at the GitHub repository <https://github.com/conrad-blucher-institute/ml-miami-compound-flood-predictions>. In the repository there exists a script that utilizes the dataretrieval module from Python (found at <https://pypi.org/project/dataretrieval/>) to download the target groundwater level time series from the USGS water data API. The rainfall and spillway parameter time series were downloaded using the online tool <https://www.sfwmd.gov/science-data/dbhydro> from SFWMD. The ocean water level time series was downloaded from the NOAA CO-OPS Tides & Currents data API (<https://api.tidesandcurrents.noaa.gov/api/prod/>) using the noaa-coops Python module (<https://pypi.org/project/noaa-coops/>).

References

- Ali, J., T. Wahl, A. R. Enriquez, M. M. Rashid, J. Morim, M. Gall, and C. T. Emrich, 2023: The role of compound climate and weather extreme events in creating socio-economic impacts in south florida. *Weather and Climate Extremes*, **42**, 100625, doi:<https://doi.org/10.1016/j.wace.2023.100625>.
- Ayus, I., N. Natarajan, and D. Gupta, 2023: Prediction of water level using machine learning and deep learning techniques. *Iranian Journal of Science and Technology, Transactions of Civil Engineering*, **47**, 2437–2447, doi:[10.1007/s40996-023-01053-6](https://doi.org/10.1007/s40996-023-01053-6).
- Breiman, L., 2001: Random forests. *Machine learning*, **45** (1), 5–32.
- Chen, S., Y. Feng, and H. Li, 2024: Enhancing runoff predictions in data-sparse regions through hybrid deep learning and hydrologic modeling. *Scientific Reports*, **14**, 26450, doi:[10.1038/s41598-024-77678-y](https://doi.org/10.1038/s41598-024-77678-y).
- Czajkowski, J., V. Engel, C. Martinez, A. Mirchi, D. Watkins, M. C. Sukop, and J. D. Hughes, 2018: Economic impacts of urban flooding in south florida: Potential consequences of managing

groundwater to prevent salt water intrusion. *Science of The Total Environment*, **621**, 465–478,
doi:<https://doi.org/10.1016/j.scitotenv.2017.10.251>.

Estrada, B., P. E. Tissot, and M. Vicens-Miquel, 2025: Machine learning based water level predictions for southern florida. <https://ams.confex.com/ams/105ANNUAL/meetingapp.cgi/Paper/455138>, Poster presented at the 105th American Meteorological Society Annual Meeting, New Orleans, Louisiana, January 12–16, 2025.

Harris, A., 2023: No-name storm swamps south florida with flooding, high winds. recovery is under way. URL <https://www.miamiherald.com/news/local/article281936518.html>, miami Herald.

Hsieh, W. W., 2023: *Introduction to Environmental Data Science*. Cambridge University Press.

Kingma, D. P., and J. Ba, 2017: Adam: A method for stochastic optimization. URL <https://arxiv.org/abs/1412.6980>, 1412.6980.

Li, W., H. Jiang, D. Li, P. B. Bedient, and Z. N. Fang, 2023: Evaluation of radar precipitation products and assessment of the gauge-radar merging methods in southeast texas for extreme precipitation events. *Remote Sensing*, **15** (8), doi:[10.3390/rs15082033](https://doi.org/10.3390/rs15082033).

Moradian, S., A. AghaKouchak, S. Gharbia, C. Broderick, and A. I. Olbert, 2024: Forecasting of compound ocean-fluvial floods using machine learning. *Journal of Environmental Management*, **364**, 121 295, doi:<https://doi.org/10.1016/j.jenvman.2024.121295>.

Morgan, N., and H. Bourlard, 1989: Generalization and parameter estimation in feedforward nets: Some experiments. *Advances in Neural Information Processing Systems*, D. Touretzky, Ed., Morgan-Kaufmann, Vol. 2, URL https://proceedings.neurips.cc/paper_files/paper/1989/file/63923f49e5241343aa7acb6a06a751e7-Paper.pdf.

Muñoz, D. F., H. Moftakhari, and H. Moradkhani, 2024: Quantifying cascading uncertainty in compound flood modeling with linked process-based and machine learning models. *Hydrology and Earth System Sciences*, **28** (11), 2531–2553, doi:[10.5194/hess-28-2531-2024](https://doi.org/10.5194/hess-28-2531-2024).

Peña, F., F. Nardi, A. Melesse, J. Obeysekera, F. Castelli, R. M. Price, T. Crawl, and N. Gonzalez-Ramirez, 2022: Compound flood modeling framework for surface–subsurface water interactions. *Natural Hazards and Earth System Sciences*, **22** (3), 775–793, doi:[10.5194/nhess-22-775-2022](https://doi.org/10.5194/nhess-22-775-2022).

- Sadler, J., J. Goodall, M. Morsy, and K. Spencer, 2018: Modeling urban coastal flood severity from crowd-sourced flood reports using poisson regression and random forest. *Journal of Hydrology*, **559**, 43–55, doi:<https://doi.org/10.1016/j.jhydrol.2018.01.044>.
- Seneviratne, S. I., and Coauthors, 2021: Weather and climate extreme events in a changing climate. *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*, V. Masson-Delmotte, P. Zhai, A. Pirani, S. L. Connors, C. Péan, S. Berger, N. Caud, Y. Chen, L. Goldfarb, M. I. Gomis, M. Huang, K. Leitzell, E. Lonnoy, J. B. R. Matthews, T. K. Maycock, T. Waterfield, O. Yelekçi, R. Yu, and B. Zhou, Eds., Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA, 1513–1766, doi:[10.1017/9781009157896.013](https://doi.org/10.1017/9781009157896.013).
- Sham, F. A. F., A. El-Shafie, and W. Z. B. W. Jaafar, 2025: Improving rainfall forecasting using deep learning data fusing model approach for observed and climate change data. *Scientific Reports*, **15**, 27 872, doi:[10.1038/s41598-025-13567-2](https://doi.org/10.1038/s41598-025-13567-2).
- Shi, J., Z. Yin, A. Leon, J. Obeysekera, and G. Narasimhan, 2025: Fiddlar: Forecast-informed deep learning architecture for flood mitigation. *Proceedings of the AAAI Conference on Artificial Intelligence*, **39** (27), 28 377–28 385, doi:[10.1609/aaai.v39i27.35059](https://doi.org/10.1609/aaai.v39i27.35059).
- Srivastava, N., G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, 2014: Dropout: a simple way to prevent neural networks from overfitting. *J. Mach. Learn. Res.*, **15** (1), 1929–1958.
- Sukop, M. C., M. Rogers, G. Guannel, J. M. Infanti, and K. Hagemann, 2018: High temporal resolution modeling of the impact of rain, tides, and sea level rise on water table flooding in the arch creek basin, miami-dade county florida usa. *Science of The Total Environment*, **616-617**, 1668–1688, doi:<https://doi.org/10.1016/j.scitotenv.2017.10.170>.
- Tang, Y., C. Zang, Y. Wei, and M. Jiang, 2019: Data-driven modeling of groundwater level with least-square support vector machine and spatial–temporal analysis. *Geotechnical and Geological Engineering*, **37** (3), 1661–1670, doi:[10.1007/s10706-018-0713-6](https://doi.org/10.1007/s10706-018-0713-6).
- Tao, H., and Coauthors, 2022: Groundwater level prediction using machine learning models: A comprehensive review. *Neurocomputing*, **489**, 271–308, doi:<https://doi.org/10.1016/j.neucom.2022.03.014>.

- 635 Terink, W., H. Leijnse, G. van den Eertwegh, and R. Uijlenhoet, 2018: Spatial resolutions in areal
636 rainfall estimation and their impact on hydrological simulations of a lowland catchment. *Journal*
637 *of Hydrology*, **563**, 319–335, doi:[10.1016/j.jhydrol.2018.05.045](https://doi.org/10.1016/j.jhydrol.2018.05.045).
- 638 Villarini, G., P. V. Mandapaka, W. F. Krajewski, and R. J. Moore, 2008: Rainfall and sam-
639 pling uncertainties: A rain gauge perspective. *Journal of Geophysical Research: Atmo-*
640 *spheres*, **113** (D11), doi:[10.1029/2007JD009214](https://doi.org/10.1029/2007JD009214), [https://agupubs.onlinelibrary.wiley.com/doi/](https://agupubs.onlinelibrary.wiley.com/doi/pdf/10.1029/2007JD009214)
641 [pdf/10.1029/2007JD009214](https://doi.org/10.1029/2007JD009214).
- 642 Wahl, T., S. Jain, and J. Bender, 2015: Increasing risk of compound flooding from
643 storm surge and rainfall for major us cities. *Nature Clim Change*, **5**, 1093–1097,
644 doi:<https://doi.org/10.1038/nclimate2736>.
- 645 Xu, K., Z. Han, L. Bin, and Coauthors, 2025: Rapid forecasting of compound flooding for a coastal
646 area based on data-driven approach. *Natural Hazards*, **121**, 1399–1421, doi:[10.1007/s11069-](https://doi.org/10.1007/s11069-024-06846-0)
647 [024-06846-0](https://doi.org/10.1007/s11069-024-06846-0).
- 648 Yadav, S., and S. Shukla, 2016: Analysis of k-fold cross-validation over hold-out validation on
649 colossal datasets for quality classification. *2016 IEEE 6th International Conference on Advanced*
650 *Computing (IACC)*, 78–83, doi:[10.1109/IACC.2016.25](https://doi.org/10.1109/IACC.2016.25).